

Sai Teja Talluri

United States | 9408437242 | sajtejatalluri99@gmail.com | <https://www.linkedin.com/in/sai-teja-t-6b7754b8/>

PROFESSIONAL SUMMARY

Senior Data Engineer with 5+ years building production-scale lakehouse platforms on AWS and Databricks across financial services, healthcare, and retail. Specialized in high-volume ETL, Delta Lake incremental pipelines, and cost-optimized Spark workloads. Proven record of cutting cloud costs by 25%, improving query performance by 20%, and delivering governed, audit-ready data systems used by analytics, ML, and compliance teams.

TECHNICAL SKILLS

- **Languages:** Python, SQL, Scala, Shell Scripting
- **Cloud & Data Platform:** AWS S3, AWS Glue, Lambda, Step Functions, Redshift, Athena, Azure Data Factory, Azure Data Lake Storage, Azure Blob Storage, Azure Synapse Analytics, Azure Databricks, Kinesis
- **Data Engineering:** Incremental Ingestion, CDC, Lakehouse Architecture, Data Modeling, Medallion architecture
- **ETL/ELT Tools:** AWS Glue, Informatica PowerCenter, Talend, Apache NiFi, DataStage
- **Big Data & Processing:** Apache Spark, PySpark, Delta Lake, Kafka, Airflow
- **Databases & Warehousing:** Snowflake, Amazon Redshift, Oracle, SQL Server, PostgreSQL, Teradata, MySQL, Azure Synapse Analytics
- **Data Modeling:** Star Schema, Snowflake Schema, Dimensional Modeling, Data Vault, ER Modeling, Data Lakehouse Architecture
- **Machine Learning & Analytics:** Scikit-learn, MLlib, AWS SageMaker, Feature Store, Data Visualization
- **Data Quality & Governance:** Great Expectations, PyDeequ, Glue Catalog, Unity Catalog, Azure Purview
- **Version Control & CI/CD:** Git, GitHub Actions, Bitbucket, Jenkins, AWS CodePipeline, Terraform, CloudFormation
- **Monitoring & Logging:** CloudWatch, Datadog, Prometheus, Grafana, Azure Monitor
- **Domains:** Financial Services (SOX, PCI), Healthcare (HIPAA), Retail Analytics

EXPERIENCE

Capital One

Aug 2024 - Present

Senior Data Engineer

Mclean, VA

- Built and scaled **Databricks lakehouse pipelines** on AWS ingesting multi-source financial data from APIs, RDS, and on-prem systems into S3 for analytics and reporting.
- Designed **Delta Lake incremental ingestion** with ACID guarantees and time travel, eliminating full reloads and improving update efficiency by **60%**.
- Implemented **Medallion architecture** (Bronze/Silver/Gold) using Delta Lake, **Delta Live Tables (DLT)**, and **Lakeflow** for batch and streaming financial data pipelines.
- Optimized Spark workloads using **partitioning, Z-ordering, caching, and cluster tuning**, improving query performance by **20%** and reducing compute costs by **25%**.
- Developed **PySpark transformation** frameworks that clean, aggregate, and join large structured and semi-structured datasets with schema consistency, ensuring data quality and **reducing downstream processing errors**.
- **Migrated legacy** ETL pipelines from **Informatica and SQL Server** to **Databricks** and AWS Glue, improving maintainability and cutting infrastructure costs.
- Orchestrated end-to-end workflows using **AWS Step Functions and Lambda**, enabling event-driven ingestion, transformation, and validation with CloudWatch monitoring.
- Automated **metadata discovery** and schema management using **AWS Glue Crawlers** and **Glue Data Catalog**, improving data governance and lineage visibility.
- Implemented **data quality validation checks** using **Great Expectations** and Pandas to detect schema drift, nulls, duplicates, and data anomalies.
- Designed and tuned **Databricks SQL** and **Athena queries** to support ad-hoc analysis and BI dashboards in Power BI and QuickSight.
- Implemented AWS IAM roles, **S3 bucket policies**, and **KMS encryption** to enforce data security and compliance with financial **regulatory standards** such as **SOX** and **PCI DSS**.
- Deployed **CI/CD** pipelines with **Jenkins** and **GitHub Actions** for Databricks notebooks and Glue jobs, enabling faster release cycles and consistent version control across the team.
- Automated data pipeline monitoring through **AWS CloudWatch Logs**, **SNS alerts**, and Datadog dashboards, reducing average incident resolution time by **35%** and **minimizing unplanned downtime**.
- Partnered with **data architects** and **analysts** to design **data models** using **Snowflake** schema for **Redshift data marts**, ensuring data quality, referential integrity, and **optimized analytical performance**.
- Supported **sprint cycles** by building **reusable ETL** components and conducting **peer code reviews**, which accelerated pipeline delivery and ensured compliance with enterprise **data governance standards**.

Sentara Healthcare

Jun 2021 - Jul 2023

Data Engineer

Bengaluru, India

- Built and operated **Databricks and AWS Glue** pipelines processing claims, EMR, and patient data for analytics and reporting teams.
- Developed **PySpark ETL workflows** to transform **HL7, FHIR**, and JSON healthcare data into structured, analytics-ready datasets.
- Implemented **CDC-based incremental ingestion** using Delta Lake, reducing data latency from hours to near-real-time.
- Orchestrated end-to-end pipelines using AWS Step Functions and Lambda, improving reliability and reducing manual reruns.
- Delivered curated datasets to **Redshift and Snowflake**, enabling clinical, financial, and operational dashboards.
- Implemented **partition pruning, caching, and broadcast joins** in Databricks SQL to **optimize Spark jobs, improving performance** of large-scale clinical data aggregations and reporting pipelines.

- Built and **optimized Redshift data marts** using **star schema** design for supporting healthcare KPIs, financial analysis, and patient utilization reporting.
- Built **data quality validation checks** using Great Expectations and PyDeequ, **cutting schema** and completeness **issues by 50%**.
- **Automated Databricks deployments** and data validation using Jenkins and GitHub Actions, **reducing release effort by 45%**.
- **Collaborated with data scientists and analysts** to deliver high-quality, curated datasets for **AI/ML healthcare models**, using **PySpark preprocessing** and feature engineering to support patient utilization and readmission models.
- Applied **IAM, encryption, masking** and access controls to ensure **HIPAA and SOC 2** compliant data ingestion and storage.
- **Monitored pipelines** using **CloudWatch and Datadog**, reducing time-to-detection by **30% for data failures**.

Sephora

Jan 2020 - May 2021

Data Engineer

Bengaluru, India

- Developed **Azure Data Factory pipelines** ingesting customer, sales, and inventory data into **Azure Data Lake Storage** and **Azure Blob Storage**, enabling unified retail analytics and faster reporting.
- Supported near **real-time ingestion** using **Azure Event Hubs** and Stream Analytics for clickstream and transaction data, improving reporting latency by 30%.
- Built **Delta Lake tables** in **Azure Databricks** to support versioned and reliable analytics datasets.
- Wrote **PySpark transformations** to cleanse and aggregate retail datasets for downstream analytics.
- Implemented metadata discovery and lineage tracking using Azure Purview and Data Catalog.
- Supported development of **Azure Synapse Analytics** data marts for customer and product analytics.
- Automated API ingestion from Salesforce and Shopify using **Azure Functions** and Logic Apps, reducing manual data loads.
- Supported **event-driven workflows** using Azure Functions, Event Grid, and **ADF triggers** for incremental refresh.
- Integrated **Power BI** with **Azure Synapse** and **Azure SQL** to deliver real-time insights on sales, inventory, and marketing KPIs
- Followed **RBAC and Key Vault security practices** to support **GDPR** and **CCPA** compliant data handling.

EDUCATION

University of North Texas

Master of Science (MS), Computer Science

Denton, Texas, USA

CERTIFICATIONS

- **AWS Certified Data Engineer: Amazon Web Services**
- **Databricks Certified Data Engineer: Databricks**
- **Microsoft Certified Fabric Data Engineer: Microsoft**